Use the force diagram to find acceleration

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Component / ES: componente / PT: componente

The downhill part of weight. A way to calculate — not an extra force.

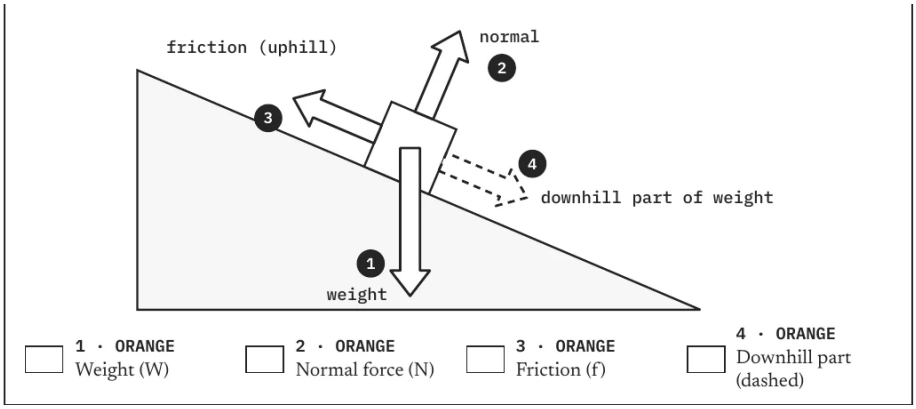
Also on your note page: Normal force, Weight, Acceleration direction.

B READ THE RELATIONSHIP

A component is the part of one vector along one chosen axis.

Resolve one force along two perpendicular axes

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration.

Given m = 2 kg 7 N downhill 3 N uphill

Rule a = F net ÷ m

Work F net = 7 − 3 = 4 N downhill a = 4 ÷ 2

Answer 2 m/s² downhill

E SAVE FOR THE FINAL MEETING

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

Read the physics: Component

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration. 2 m/s² downhill.

05 CONNECT TO XJ

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Normal force

ES: fuerza normal / PT: força normal

Meaning in my words	How can you tell?
How would you explain normal force to a friend? In my words, this means ___.	A crate rests on a ramp. Which way does the ramp push the crate? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw the normal force on a book on a level table? My example shows ___.	Someone draws the normal force along the ramp. What would you rotate? I would change ___ because ___.

Weight

ES: peso / PT: peso

Meaning in my words	How can you tell?
How would you explain weight to a friend? In my words, this means ___.	When a book rests on a table, which object supplies its weight? I can tell because ___.
Example or sketch	**Non-example and why**
Using 10 N/kg, what arrow represents the weight of a 3 kg book? My example shows ___.	A book slides down a ramp. Should its whole weight arrow point along the ramp? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Component

ES: componente / PT: componente

Meaning in my words	**How can you tell?**
How would you explain component to a friend? In my words, this means ___.	Which part of the block's weight pulls it down the ramp? I can tell because ___.
Example or sketch	**Non-example and why**
How could you split one downward weight arrow using ramp axes? My example shows ___.	Should the downhill component be added to the full weight as another force? I would change ___ because ___.

Acceleration direction

ES: sentido de la aceleración / PT: sentido da aceleração

Meaning in my words	**How can you tell?**
How would you explain acceleration direction to a friend? In my words, this means ___.	A cart rolls uphill on a frictionless ramp. Which way does it accelerate? I can tell because ___.
Example or sketch	**Non-example and why**
At the top of its trip, draw the acceleration arrow just as the cart stops. My example shows ___.	"The cart moves uphill, so acceleration points uphill." Which arrows got mixed up? I would change ___ because ___.

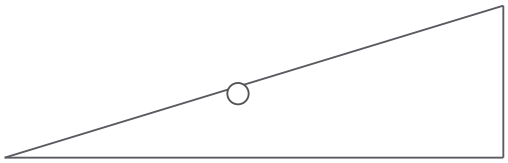
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

Tell a partner

Along the ramp I use ___. Across the ramp I use ___.

On assignment: Component

SECTION E / BACK / FINAL MEETING ONLY Name _____

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

Word bank: parallel · perpendicular · downhill.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”